NAMAL UNIVERSITY MIANWALI

Department of Computer Science

Database Design Document

CCP - Milestone 2

CattleCare Pro: Livestock Breeding and Farm Management Database System

Subject: CSC-271 – Database Systems

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Chapter 1: Project Overview

1.1 Introduction

Livestock farming constitutes a significant pillar of the agricultural economy in Pakistan and across many developing nations. Effective management of cattle records, breeding history, and financial transactions data is essential for sustainable and profitable farm operations. Despite this importance, the majority of livestock farms continue to rely on manual record-keeping practices, including paper registers, handwritten ledgers, and basic spreadsheets. These traditional approaches introduce numerous inefficiencies: data loss due to physical damage or misplacement, inconsistency caused by human error, and extreme difficulty in retrieving or analyzing historical records.

CattleCare Pro is a proposed relational database-driven system designed specifically for livestock breeding and farm management. The system aims to replace fragmented manual methods with a structured, normalized, and query-efficient SQL database. It will maintain comprehensive records of cattle identification, breeding lineage, birth and death events, purchase and sale transactions, periodic weight and color tracking, and financial profit or loss calculations. By centralizing all farm data into a single coherent relational schema, CattleCare Pro will enable farm owners to make evidence-based decisions, trace complete animal histories, and manage their operations with greater precision and accountability.

The system is built around a carefully designed Enhanced Entity-Relationship (EERD) model that captures both the common attributes of all cattle and the specialized attributes of different cattle subtypes, namely cows and bulls. This EERD forms the foundation of the database schema to be implemented in MySQL, ensuring referential integrity, minimal redundancy, and scalability as farm data grows over time.

1.2 Problem Statement

Many livestock farm operators face serious challenges when relying on manual or informal systems for cattle management. The core problems identified are as follows:

- Difficulty maintaining complete and accurate cattle identification records across large herds.
- Lack of organized and searchable breeding history, making genetic lineage tracking nearly impossible.
- Inability to efficiently track birth events and link newborns to their parents in a structured manner.
- Poor management of cattle death records, resulting in outdated inventory and incorrect financial calculations.
- Disorganized purchase and sale transaction logs that complicate auditing and profit determination.
- Absence of systematic weight and color tracking, which are important indicators for breeding selection and market valuation.
- No structured mechanism to compute and record profit or loss per animal after sale.
- Inefficient retrieval of historical records when required for inspections, loans, or insurance

claims.

The cumulative effect of these problems results in poor financial performance visibility, breeding decisions based on incomplete information, and an overall inability to scale farm operations. A properly designed relational database system is therefore essential to address these limitations systematically.

1.3 Project Objectives

The primary objective of CattleCare Pro is to design and implement a normalized relational SQL database that enables efficient storage, retrieval, and management of all livestock breeding and farm management data.

Specific objectives include:

- Design a fully normalized relational database schema for comprehensive livestock data management.
- Store detailed cattle identification records including tag number, breed, sex, date of birth, and current status.
- Implement a supertype–subtype structure to differentiate cows and bulls while sharing common cattle attributes.
- Maintain complete breeding records linking cows and bulls to each breeding event.
- Track birth records and associate each newborn with its breeding event.
- Record death information for cattle that expire, preserving historical inventory accuracy.
- Manage purchase and sale transactions with complete pricing information.
- Store periodic weight and color observations for each animal.
- Automatically compute and store profit or loss figures derived from purchase and sale records.

GitHub Repository: `https://github.com/aligoharcodepulse/CattleCare-Pro/`

Chapter 2: Detailed Database Design

2.1 Entity Identification

The following table identifies all entities present in the CattleCare Pro system. Each entity represents a distinct real-world object or concept that requires persistent storage within the database. The *Cattle* entity serves as a supertype, with *Cow* and *Bull* as its subtypes under an exclusive, total specialization.

Table 1. Entity Identification

Sr. No.	Entity Name	Description
01	User	Farm administrator or operator who has login credentials and manages all farm data.
02	Farm	Physical farm or livestock unit registered in the system, owned and managed by a user.
03	Cattle	Supertype entity representing any animal in the farm, storing common attributes shared by all cattle.
04	Cow	Subtype of Cattle. Represents female cattle with attributes specific to their reproductive role.
05	Bull	Subtype of Cattle. Represents male cattle with attributes specific to their role as sires in breeding.
06	Breeding Record	Records a breeding event between a cow and a bull, capturing when and how the breeding occurred.
07	Birth Record	Records the birth of a calf, linking the newborn to its breeding event.
08	Death Record	Stores mortality information for cattle that have died, including date and cause.
09	Purchase Record	Documents the purchase of a cattle animal, capturing vendor details, date, and purchase price.
10	Sale Record	Documents the sale of a cattle animal, capturing buyer details, date, and sale price.
11	Weight Record	Stores periodic weight measurements for a cattle animal to track growth over time.
12	Color Record	Records color observations of a cattle animal used for identification and breeding selection.
13	Profit/Loss Record	Derived financial record that computes and stores the profit or loss from the sale of an animal.

2.2 Data Dictionary

The data dictionary below provides a comprehensive definition of all attributes for each entity. It specifies attribute names, data types, constraints, and descriptive notes.

User

Table 2. User Entity Attributes

Sr.	Attribute	Data Type	Constraint	Description
01	UserID *(PK)*	INT	NOT NULL, AUTO_INC, PK	Unique system-generated identifier for each user account.
02	FullName	VARCHAR	NOT NULL	Full legal name of the user.
03	Email	VARCHAR	NOT NULL, UNIQUE	Email address used for login. Must be unique across all users.
04	Password	VARCHAR	NOT NULL	Hashed password string for secure authentication.
05	Role	ENUM	NOT NULL, DEFAULT ‘admin’	Role of the user: ‘admin’ or ‘viewer’.
06	Phone	VARCHAR	NULL	Optional contact phone number for the user.

Farm

Table 3. Farm Entity Attributes

Sr.	Attribute	Data Type	Constraint	Description
01	FarmID *(PK)*	INT	NOT NULL, AUTO_INC, PK	Unique identifier for each registered farm.
02	FarmName	VARCHAR	NOT NULL	Name of the farm as registered in the system.
03	Location	VARCHAR	NOT NULL	Physical address or geographic location of the farm.
04	Size	DECIMAL	NULL	Total area of the farm in acres or hectares.
05	OwnerID *(FK)*	INT	NOT NULL	Contact Info of the Farm Owner.

Cattle (Supertype)

Table 4. Cattle Supertype Entity Attributes

Sr.	Attribute	Data Type	Constraint	Description
01	CattleID *(PK)*	INT	NOT NULL, AUTO_INC, PK	Unique identifier assigned to each animal upon registration.
02	TagNumber	VARCHAR	NOT NULL, UNIQUE	Physical ear-tag number visible on the animal.
03	Breed	VARCHAR	NOT NULL	Recognized breed name (e.g., Sahiwal, Frisian, Cholistani).
04	Gender	ENUM	NOT NULL	Biological sex: ‘M’ for male (bull) or ‘F’ for female (cow).
05	DateOfBirth	DATE	NOT NULL	Recorded or estimated birth date of the animal.
06	Status	ENUM	NOT NULL, DEFAULT ‘active’	Current life status: ‘active’, ‘sold’, or ‘dead’.
07	FarmID *(FK)*	INT	NOT NULL	To which farm the cattle belongs.

Cow (Subtype)

Table 5. Cow Subtype Entity Attributes

Sr.	Attribute	Data Type	Constraint	Description
01	CattleID(CowID) *(PK/FK)*	INT	NOT NULL, PK, FK → CattleID	Inherits CattleID. Serves as both primary and foreign key.
02	LactationStatus	ENUM	NOT NULL, DEFAULT ‘dry’	Current milk production status: ‘milking’ or ‘dry’.
03	LastCalvingDate	DATE	NULL	Date of the most recent calving event for this cow.

Bull (Subtype)

Table 6. Bull Subtype Entity Attributes

Sr.	Attribute	Data Type	Constraint	Description
01	BullID *(PK/FK)*	INT	NOT NULL, PK, FK → CattleID	Inherits CattleID. Serves as both primary and foreign key.
02	FertilityStatus	VARCHAR	NOT NULL	Indicates the fertility condition of the bull (e.g., High, Moderate, Low).
03	SemenQuality	VARCHAR	NOT NULL	Represents the quality grade of semen used for breeding purposes.

Breeding Record

Table 7. Breeding Record Entity Attributes

Sr.	Attribute	Data Type	Constraint	Description
01	BreedingID *(PK)*	INT	NOT NULL, AUTO_INC, PK	Unique identifier for each breeding event.
02	CowID *(FK)*	INT	NOT NULL, FK	References the cow involved in the breeding event.
03	BullID *(FK)*	INT	NOT NULL, FK	References the bull used for breeding.
04	BreedingDate	DATE	NOT NULL	Calendar date on which the breeding event occurred.
05	Method	ENUM	NOT NULL	Breeding method used: ‘natural’ or ‘AI’ (Artificial Insemination).
06	ExpectedDueDate	DATE	NULL	Estimated due date for the cow after successful breeding.
07	Outcome	VARCHAR(50)	NULL	Result of breeding such as ‘Pregnant’, ‘Failed’, or ‘Pending’.
08	Notes	TEXT	NULL	Optional veterinary or farmer notes regarding the breeding event.

Birth Record

Table 8. Birth Record Entity Attributes

Sr.	Attribute	Data Type	Constraint	Description
01	BirthID *(PK)*	INT	NOT NULL, AUTO_INC, PK	Unique identifier for each birth event.
02	MotherID *(FK)*	INT	NOT NULL, FK	References the mother cow that gave birth to the calf.
03	CalfID *(FK)*	INT	NOT NULL, FK	References the newborn calf associated with the birth record.
04	BreedingID *(FK)*	INT	NOT NULL, FK	References the breeding event linked to this birth.
05	BirthDate	DATE	NOT NULL	Date on which the calf was born.
06	BirthWeight	DECIMAL	NULL	Weight of the calf at birth in kilograms.
07	Complications	TEXT	NULL	Notes on any birth complications observed during delivery.

Death Record

Table 9. Death Record Entity Attributes

Sr.	Attribute	Data Type	Constraint	Description
01	DeathID *(PK)*	INT	NOT NULL, AUTO_INC, PK	Unique identifier for each death record.
02	CattleID *(FK)*	INT	NOT NULL, FK	References the cattle animal associated with the death record.
03	DeathDate	DATE	NOT NULL	Date on which the cattle animal died.
04	Cause	VARCHAR(	NULL	Recorded cause of death such as disease, injury, or old age.
05	VetVerified	BOOLEAN	NOT NULL, DEFAULT FALSE	Indicates whether the death record has been verified by a veterinarian.

Purchase Record

Table 10. Purchase Record Entity Attributes

Sr.	Attribute	Data Type	Constraint	Description
01	PurchaseID *(PK)*	INT	NOT NULL, AUTO_INC, PK	Unique identifier for each cattle purchase record.
02	CattleID *(FK)*	INT	NOT NULL, FK	References the purchased cattle animal in the CATTLE table.
03	UserID *(FK)*	INT	NOT NULL, FK	References the user who recorded or managed the purchase transaction.
04	SellerName	VARCHAR	NOT NULL	Name of the person or vendor from whom the cattle was purchased.
05	PurchaseDate	DATE	NOT NULL	Date on which the cattle animal was purchased.
06	PurchasePrice	DECIMAL	NOT NULL	Amount paid for purchasing the cattle animal, in Pakistani Rupees.
07	MarketName	VARCHAR	NULL	Name of the livestock market or location where the purchase took place.

Sale Record

Table 11. Sale Record Entity Attributes

Sr.	Attribute	Data Type	Constraint	Description
01	SaleID *(PK)*	INT	NOT NULL, AUTO_INC, PK	Unique identifier for each cattle sale transaction.
02	CattleID *(FK)*	INT	NOT NULL, FK	References the cattle animal sold from the CATTLE table.
03	UserID *(FK)*	INT	NOT NULL, FK	References the user who recorded or managed the sale transaction.
04	BuyerName	VARCHAR	NOT NULL	Name of the person or entity who purchased the cattle animal.
05	SaleDate	DATE	NOT NULL	Date on which the cattle animal was sold.
06	SalePrice	DECIMAL	NOT NULL	Amount received from selling the cattle animal, in Pakistani Rupees.
07	MarketName	VARCHAR	NULL	Name of the livestock market or location where the sale took place.

Weight Record

Table 12. Weight Record Entity Attributes

Sr.	Attribute	Data Type	Constraint	Description
01	WeightID *(PK)*	INT	NOT NULL, AUTO_INC, PK	Unique identifier for each weight measurement record.
02	CattleID *(FK)*	INT	NOT NULL, FK	References the cattle animal whose weight is being recorded.
03	WeightDate	DATE	NOT NULL	Date on which the animal's weight was measured.
04	WeightKg	DECIMAL	NOT NULL	Weight of the cattle animal in kilograms at the time of measurement.

Color Record

Table 13. Color Record Entity Attributes

Sr.	Attribute	Data Type	Constraint	Description
01	ColorID *(PK)*	INT	NOT NULL, AUTO_INC, PK	Unique identifier for each cattle color record.
02	CattleID *(FK)*	INT	NOT NULL, FK	References the cattle animal whose color details are being recorded.
03	PrimaryColor	VARCHAR	NOT NULL	Main coat color of the cattle animal (e.g., white, black, brown).
04	SecondaryColor	VARCHAR	NULL	Additional coat color if the animal has mixed coloring or markings.
05	Pattern	VARCHAR	NULL	Description of the coat pattern such as spotted, solid, or patchy.

Profit/Loss Record

Table 14. Profit/Loss Record Entity Attributes

Sr.	Attribute	Data Type	Constraint	Description
01	RecordID *(PK)*	INT	NOT NULL, AUTO_INC, PK	Unique identifier for each profit or loss record.
02	CattleID *(FK)*	INT	NOT NULL, FK	References the cattle animal associated with the financial calculation.
03	PurchaseID *(FK)*	INT	NOT NULL, FK	References the related purchase transaction record.
04	SaleID *(FK)*	INT	NOT NULL, FK	References the related sale transaction record.
05	ProfitLoss	DECIMAL	NOT NULL	Net profit or loss calculated from the cattle transaction.
06	CalculationDate	DATE	NOT NULL	Date on which the profit or loss was calculated.

2.3 Business Rules

The following business rules define the operational constraints and behavioral policies enforced by the CattleCare Pro database system.

BR1. A **Farm** must be associated with exactly one **User** (administrator). A User may manage one or more Farms.

BR2. A **Farm** must house at least one **Cattle** animal at any point in time. Each Cattle animal belongs to exactly one Farm.

BR3. Every **Cattle** animal must be classified as either a **Cow** or a **Bull**. The specialization is exclusive and total – an animal cannot belong to both subtypes simultaneously.

BR4. A **Cattle** animal may have at most one **Death Record**. Once a death is recorded, the animal's Status must be updated to 'dead'.

BR5. A **Cattle** animal may have at most one **Purchase Record**, reflecting the initial acquisition of the animal.

BR6. A **Cattle** animal may have at most one **Sale Record**. Once sold, the animal's Status must be updated to 'sold'.

BR7. A **Cattle** animal may have multiple **Weight Records** over its lifetime.

BR8. A **Cattle** animal may have multiple **Color Records** over its lifetime.

BR9. A **Cow** may participate in zero or more **Breeding Records** as the dam. A **Bull** may participate in zero or more Breeding Records as the sire.

BR10. Each **Breeding Record** must be linked to exactly one **Cow** and exactly one **Bull**.

BR11. A **Breeding Record** may result in at most one **Birth Record**. Not every breeding event leads to a successful birth.

BR12. A **Birth Record** must be linked to exactly one **Breeding Record** and must produce exactly one new Cattle record representing the calf.

BR13. Each **Profit/Loss Record** must be associated with exactly one Cattle animal and is computed only when both a Purchase Record and a Sale Record exist.

BR14. A **Purchase Record** may contribute to at most one **Profit/Loss Record**. Similarly, a Sale Record may contribute to at most one Profit/Loss Record.

BR15. The ProfitLossAmount is calculated as:

$$\text{ProfitLossAmount} = \text{SalePrice} - \text{PurchasePrice} \tag{1}$$

A positive value represents profit; a negative value represents a loss.

BR16. A **User** must have a unique email address. No two users may share the same email.

BR17. The **TagNumber** of every Cattle animal must be unique across the entire system.

2.4 Entity Relationship Diagram

The Enhanced Entity-Relationship Diagram (EERD) below represents the complete structural design of the CattleCare Pro database. The diagram uses Crow's Foot notation to illustrate all entities, their attributes, the specialization hierarchy (Cattle → Cow / Bull), and all relationships with their cardinalities.

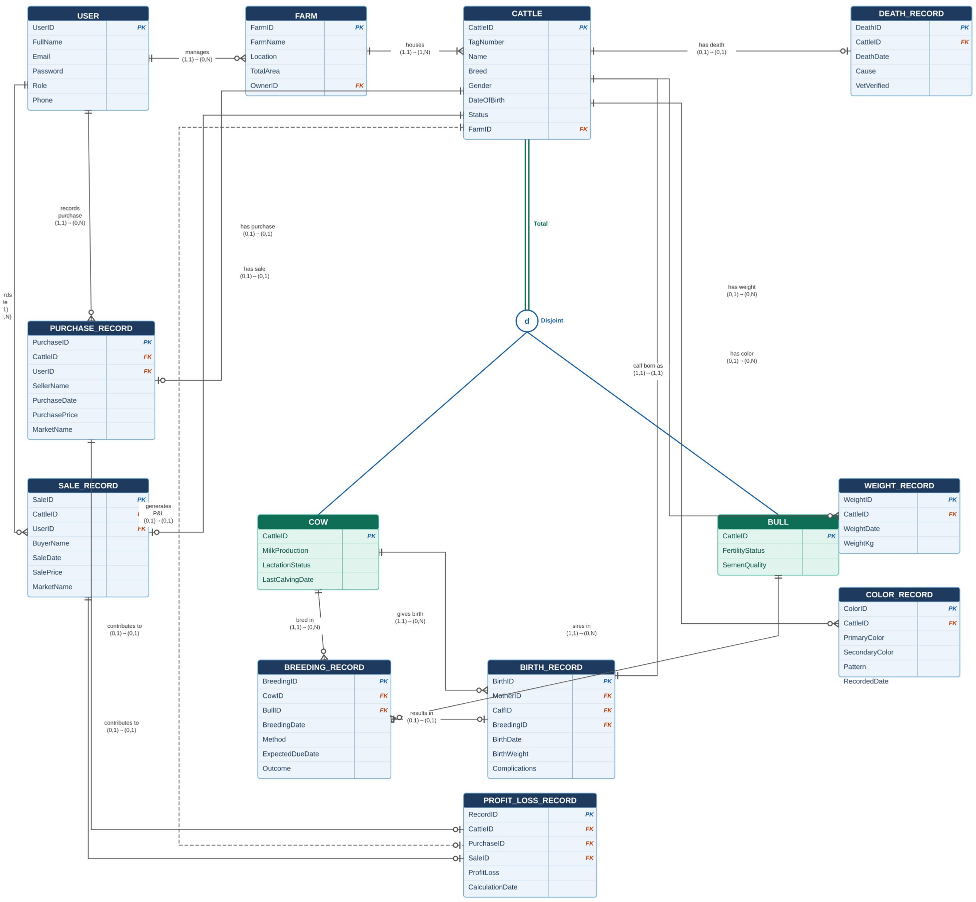

Figure 1. Enhanced Entity-Relationship Diagram of CattleCare Pro using Crow's Foot Notation.

The following table summarizes all relationships and cardinalities as represented in the EERD.

Table 15. Relationship Summary with Cardinalities

From Entity	Relationship Name	To Entity	Cardinality
User	manages	Farm	(1,1) to (0,N)
Farm	houses	Cattle	(1,1) to (1,N)
Cattle	has death	Death Record	(1,1) to (0,1)
Cattle	has purchase	Purchase Record	(1,1) to (0,1)
Cattle	has sale	Sale Record	(1,1) to (0,1)
Cattle	has weight	Weight Record	(1,1) to (0,N)
Cattle	has color	Color Record	(1,1) to (0,N)
Cattle	generates	Profit/Loss Rec.	(1,1) to (0,1)
Cow	bred in	Breeding Record	(1,1) to (0,N)
Bull	sires in	Breeding Record	(1,1) to (0,N)
Breeding Record	results in	Birth Record	(1,1) to (0,1)
Cow	gives birth to	Birth Record	(1,1) to (0,N)
Birth Record	calf born as	Cattle	(1,1) to (1,1)
Purchase Rec.	contributes to	Profit/Loss Rec.	(1,1) to (0,1)
Sale Record	contributes to	Profit/Loss Rec.	(1,1) to (0,1)

References

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[3] IEEE, "IEEE Standard for Software Requirements Specifications," *IEEE Std 830-1998*, 1998.

[4] CattleCare Pro GitHub Repository. [Online]. Available: https://github.com/aligoharcodepulse/CattleCare-Pro/

Appendix: AI Usage and Prompts

In accordance with the course guidelines, the following section documents all AI tools and online services used during the preparation of this Milestone 2 report and the development of the EERD. All AI-generated content was critically reviewed, verified against course material, and adapted by team members to ensure accuracy.

Sr.	Tool Used	Prompt / Action	Purpose
01	Claude AI	"Design a complete Enhanced Entity-Relationship Diagram (EERD) for a livestock breeding and farm management system named CattleCare Pro. The system should manage cattle records, breeding history, birth and death events, purchase and sale transactions, weight and color tracking, and financial profit or loss calculations. Use Crow's Foot notation and include all entities, attributes, relationships, and cardinalities. Deliver the output as an HTML file so that i can download it."	Used to design and generate the complete EERD for the CattleCare Pro system in HTML format, incorporating all required entities and relationships with Crow's Foot notation.
02	CloudConvert	Uploaded the HTML file containing the EERD generated by Claude and converted it to JPG image format using the HTML to JPG conversion service.	Used to convert the Claude-generated HTML-based EERD diagram into a JPG image suitable for inclusion in the project report and submission documents.
03	Claude AI	"Generate a complete Milestone 2 Database Design Document for CattleCare Pro following the provided university template. The document must include an introduction, problem statement, project objectives, entity identification table, data dictionary for all 13 entities with attributes and constraints, business rules, EERD section with relationship summary table, references, and an AI usage appendix. Format it in LaTeX following IEEE standards."	Used to generate the full structured Milestone 2 report in LaTeX, covering all required sections as specified in the assignment template provided by the instructor.
04	Claude AI	"The document has the following issues: the cover page heading order does not match the provided template, the Namal University logo is missing, and the Appendix section for AI usage is not included. Please correct these issues while keeping all other content unchanged."	Used to refine and correct specific formatting and structural issues in the LaTeX document to ensure alignment with the university-provided template and submission requirements.